DEVOPS & MLOPS FINAL PROJECT

Telecom Churn MLOps Pipeline

End-to-End ML System Development and Deployment

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COURSE
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Project Abstract

Implementation of a complete MLOps pipeline for telecom customer churn prediction. Featuring automated data versioning, experiment tracking, CI/CD, and model serving via FastAPI.

97.7% Best Model AUC

85.6% Precision Rate

78.3% Recall Rate

The Business Challenge

The Churn Crisis

Customer attrition costs telecom providers millions in lost revenue. High-value customers leaving the service significantly impacts profitability.

- ✓ Manual model development is time-consuming and error-prone
- ✓ Data drift degrades performance over time
- ✓ Production deployment needs scalable integration
- ✓ Monitoring is essential for ongoing reliability

Goal:

An automated, scalable system to predict customer churn using machine learning, enabling proactive retention strategies.

Core MLOps Principles Applied

DevOps Sync

Automated testing with Pytest, Containerization with Docker and docker-compose, and Git feature-branch workflows.

CI/CD Mastery

GitHub Actions for automated linting (flake8, black, mypy), security scans, and deployment staging with manual approval.

ML Excellence

DVC 3.49.0 for data versioning, MLflow 2.8.0 for experiment tracking, and model registry management.

Data Processing Pipeline

01. Loading
Import 50,000 high-value customer records (Top 70th percentile).

02. Features
Rolling averages, usage decay, and recharge frequency patterns.

03. SMOTE
Synthetic oversampling to balance 8.7% churn minority.

04. PCA
Reduce 114 features down to 28 components (95% variance).

05. Split
80/20 Stratified split for accurate model validation.

Fully tracked and automated via **DVC** (Data Version Control) dvc.yaml

Model Architectures

Architecture	Key Hyperparameters	AUC Score
Gradient Boosting	learning_rate=0.1, n_estimators=100	**0.9776**
Logistic Regression	C=1.0, penalty=l2, class_weight=balanced	0.9761
Random Forest	max_depth=10, min_samples_split=5, n_estimators=100	0.9664

Best Model Selection: Gradient Boosting was selected based on the highest AUC score, achieving an F1 of 0.8175.

Confusion Matrix: Gradient Boosting

Test Set Error Analysis (6,003 samples)

- ✓ **True Negatives (5,420):** Correctly identified retained customers (90.3%).
- ✓ **True Positives (403):** Correctly identified at-risk churners.
- ✓ **False Positives (68):** Incorrectly flagged (minor retention cost waste)
- ✓ **False Negatives (112):** Missed actual churners

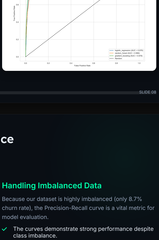
Specificity: 98.76%
Sensitivity (Recall): 78.25%

ROC Curve Analysis

Discriminative Ability

The Receiver Operating Characteristic (ROC) curves demonstrate excellent discriminative ability across all models.

Model	AUC
Gradient Boosting	0.9776
Logistic Regression	0.9761
Random Forest	0.9664



Precision-Recall Performance

Handling Imbalanced Data

Because our dataset is highly imbalanced (only 8.7% churn rate), the Precision-Recall curve is a vital metric for model evaluation.

- ✓ The curves demonstrate strong performance despite class imbalance.
- ✓ **Gradient Boosting** achieves the highest Average Precision (AP) score of **0.797**.
- ✓ Baseline AP is only 0.086.

ROI & Business Impact

Assuming 100,000 base customers | $50 ARPU | $10 Retention Cost | 8.7% Churn

+$352k
Net ROI Improvement

Metric	Without Model	With Model
Churners Identified	0	6,808
Campaign Cost	$87,000	$74,763
Revenue Saved	$0	$340,400

System Architecture & Deployment

Production Stack

- ✓ **Serving:** FastAPI for low-latency endpoints (<10ms target)
- ✓ **Containers:** Docker Multi-stage builds (~850MB Production API)
- ✓ **Cloud (AWS):** ECS Fargate, ECR, S3, Application Load Balancer

Monitoring Strategy

Retraining Triggers:

- AUC drops below 0.85
- Data drift detected (KS test $p < 0.05$)
- 30 days since last training

Conclusion & Questions

The project successfully demonstrates an industry-standard MLOps lifecycle, delivering high predictive accuracy while maintaining scalable and reproducible infrastructure.

github.com/Som21200/telecom-churn-mlops

huggingface.co/Som21200/telecom-churn-model